

0. Definitions *coordinates, the slope a'_{hd} , and the closed-loop tracking error*

$$h[k] = h_d[k] - \delta h[k], \quad \Delta h_d[k] = h_d[k+1] - h_d[k]$$

$$a'_{hd}[k] := \left. \frac{da_h}{dh} \right|_{h_d[k]}, \quad a_r[k] := a_h[k] - a_h(h_d[k])$$

$$\delta h[k+1] = \lambda_c \delta h[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k]$$

a_h = actual-height gain (state 4); a'_{hd} = its slope (state 5); a_r = residual gain from the tracking error. $a_{var} = a_{cov} = 0.05$ (EWMA weights); $\beta = 1 - T_s/\tau$ ($\tau = 0.35\text{ s} \Rightarrow \beta = 0.99821$) = the a' smoothing weight. $\delta h_1 := \delta h[k-d]$.

1. True *the residual is $-a'_{hd} \delta h$; hence the var-ratio recovers a'_{hd} , and the β -EWMA is its update law*

First-order Taylor of the gain at the perturbed height:

$$a_h(h[k]) \approx a_h(h_d[k]) - a'_{hd}[k] \delta h[k]$$

$$\boxed{a_r[k] = -a'_{hd}[k] \delta h[k], \quad a'_{hd}[k+1] \approx a'_{hd}[k]}$$

Because the residual and the tracking error share the author a'_{hd} , their variance ratio recovers the slope:

$$a'_{hdm}[k] := +\sqrt{\text{Var}(a_r[k]) / \text{Var}(\delta h[k])} = a'_{hd}[k] \quad (a'_{hd} > 0)$$

The true update law for the slope is a β -EWMA of this readout — consistent because $a'_{hdm} \equiv a'_{hd}$:

$$\boxed{a'_{hd}[k+1] = \beta a'_{hd}[k] + (1 - \beta) a'_{hdm}[k], \quad a'_{hdm} \equiv a'_{hd}}$$

2. $\hat{a}'_{hdm}$ *the estimator's empirical var-ratio: two EWMA chains on $\hat{a}_h$ and $\delta \hat{h}_3$*

Replace the true quantities by their estimates. One chain tracks the mean and variance of the gain estimate, the other of the tracking-error estimate:

$$\bar{a}_h[k+1] = (1 - a_{var}) \bar{a}_h[k] + a_{var} \hat{a}_h[k+1], \quad \hat{a}_r[k] = \hat{a}_h[k] - \bar{a}_h[k]$$

$$\hat{\sigma}_{\hat{a}_r}^2[k+1] = (1 - a_{cov}) \hat{\sigma}_{\hat{a}_r}^2[k] + a_{cov} \hat{a}_r^2[k+1]$$

$$\overline{\delta \hat{h}}[k+1] = (1 - a_{var}) \overline{\delta \hat{h}}[k] + a_{var} \delta \hat{h}_3[k+1], \quad \delta \hat{h}_r[k] = \delta \hat{h}_3[k] - \overline{\delta \hat{h}}[k]$$

$$\hat{\sigma}_{\delta \hat{h}}^2[k+1] = (1 - a_{cov}) \hat{\sigma}_{\delta \hat{h}}^2[k] + a_{cov} \delta \hat{h}_r^2[k+1]$$

$$\boxed{\hat{a}'_{hdm}[k] := +\sqrt{\hat{\sigma}_{\hat{a}_r}^2[k] / \hat{\sigma}_{\delta \hat{h}}^2[k]}}$$

3. Esti (co-update) a'_{hd} is updated by the β -EWMA of $\hat{a}'_{hdm}$ plus the KF innovation correction

State $x = [\delta h_1, \delta h_2, \delta h_3, a_h, a'_{hd}]^\top$. The KF proper corrects $[\delta h_1, \delta h_2, \delta h_3, a_h]$ from $y_1 = \delta x_m$ and $y_2 = a_{xm}$; innovations $e_{y_1} = y_1 - \hat{y}_1$, $e_{y_2} = y_2 - \hat{y}_2$; gain $K = P^- H^\top S^{-1}$, $S = H P^- H^\top + R$.

$$\hat{a}_h[k+1] = \hat{a}_h[k] + \hat{a}'_{hd}[k](\Delta h_d[k] + (1 - \lambda_c) \delta \hat{h}_3[k]) \quad (\text{gain-level predict})$$

$$\hat{a}'_{hd}[k+1] = \underbrace{\beta \hat{a}'_{hd}[k] + (1 - \beta) \hat{a}'_{hdm}[k]}_{\text{(i) var-ratio } \beta\text{-EWMA (}\S 1 \text{ form)}} + \underbrace{L_{51}[k] e_{y_1}[k] + L_{52}[k] e_{y_2}[k]}_{\text{(ii) KF } L \cdot e \text{ correction}}$$

$L_{5j} = [P^- H^\top S^{-1}]_{5,j}$. With $H(\cdot, 5) = 0$ (no own measurement row for a'_{hd}), L_{5j} is a *cross-covariance* gain $\propto P_{5,4}^- H(j, 4)$, built by the coupling $F_e(4, 5) = \Delta h_d$. Setting $L \cdot e = 0$ recovers the pure esti bolt-on; dropping $(1 - \beta) \hat{a}'_{hdm}$ recovers the pure as-state.

4. Error how the estimate errors evolve

$$e_{a_h} = a_h - \hat{a}_h, \quad e_{a'_{hd}} = a'_{hd} - \hat{a}'_{hd}, \quad e_h = \delta h - \delta \hat{h}.$$

$$e_{a_h}[k+1] = (1 + \hat{a}'_{hd} F_{dh}) e_{a_h} + \Delta h_d e_{a'_{hd}} + (1 - \lambda_c) \hat{a}'_{hd} e_h + \hat{a}'_{hd} \varepsilon_h$$

$$e_{a'_{hd}}[k+1] = \beta e_{a'_{hd}}[k] - (1 - \beta)(\hat{a}'_{hdm}[k] - a'_{hdm}[k]) - (\text{KF correction})$$

The middle term is the var-ratio readout error (self-copy + noise); the last is the genuine measurement pull — the $L \cdot e$ term of §3, kept symbolic here.

5. F_e and H the 5×5 transition and 2×5 measurement Jacobians

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{a}'_{hd}[k] & 1 + \hat{a}'_{hd}[k] F_{dh}[k] & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & \beta \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Row-5 pole = β (the EWMA memory); the $(1 - \beta) \hat{a}'_{hdm}$ drive and the $L \cdot e$ correction enter as inputs, not through the pole. $F_e(4, 5) = \Delta h_d$ is the *only* path giving a'_{hd} a Kalman gain.

6. Q and R process and measurement noise

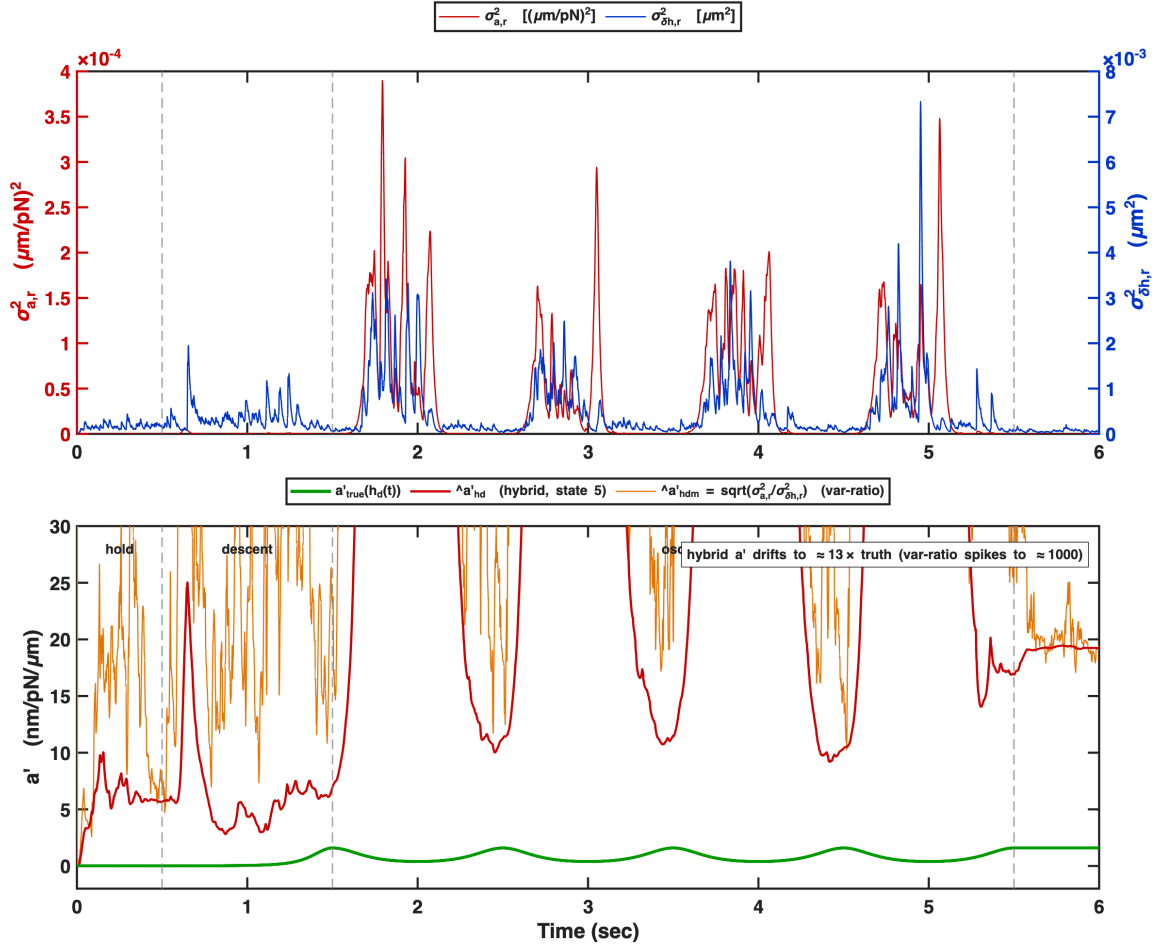
$$Q = \text{diag}(0, 0, Q_{33}, Q_{44}, Q_{55}), \quad R = \text{diag}(\sigma_n^2, R_2)$$

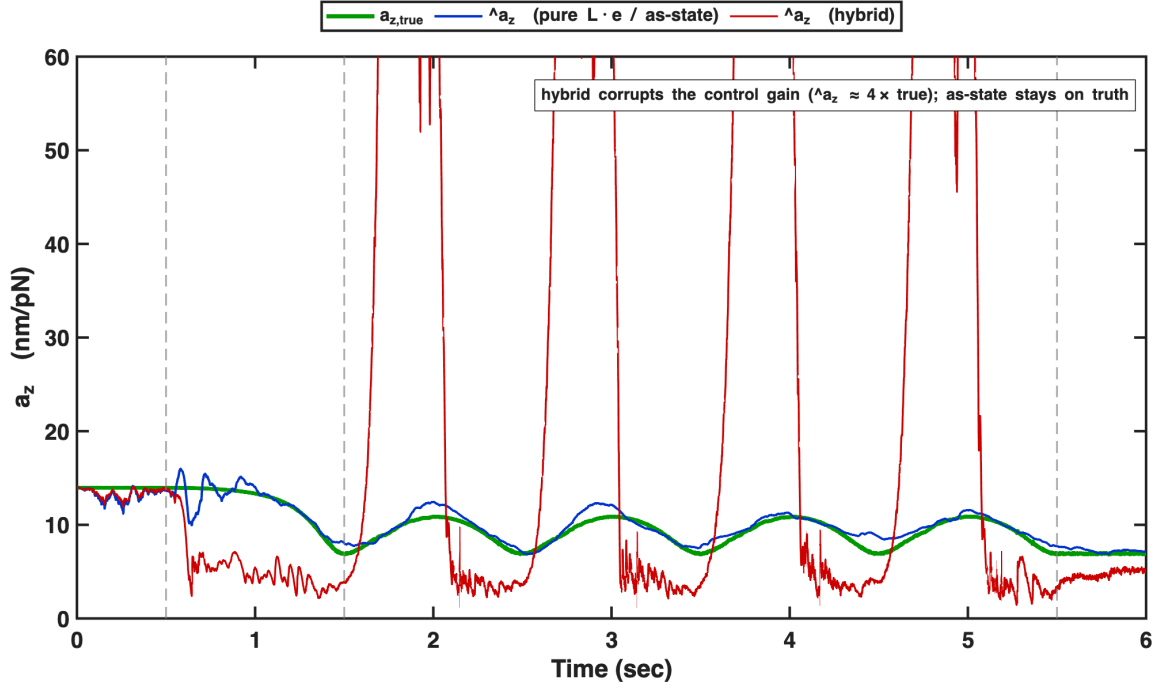
$$Q_{33} = 4k_B T \hat{a}_h (1 + (1 - \lambda_c)^2 d) + (1 - \lambda_c)^2 \sigma_n^2, \quad Q_{44} = (\hat{a}_h K_h / R)^2 \sigma_{\delta h}^2, \quad Q_{55} = (1 - \beta)^2 R_{var}$$

$$R_2 = a_{cov} IF_{var} (\hat{a}_h + \xi)^2, \quad R_{var} \approx \frac{a_{cov}}{4} (IF_{\hat{a}_r} + IF_{\delta \hat{h}} - 2IF_{\times}) \hat{a}'_{hd}{}^2$$

$\sigma_{\delta h}^2 = 4k_B T \hat{a}_h$; $\xi = (C_n / C_{dpmr}) \sigma_n^2 / (4k_B T)$; $IF_* = 1 + 2 \sum_{j \geq 1} \rho_*(j) (1 - a_{cov})^j$. Q_{55} is the injected variance of the $(1 - \beta) \hat{a}'_{hdm}$ drive.

7. Simulation (1 Hz) *co-update traces, init 1×*





8. As-state (the blue line): drop the var-ratio a'_{hd} is a plain KF state, corrected by $L \cdot e$ only — no $\hat{a}'_{hdm}$

Start from the same True relation (§1): $a_r[k] = -a'_{hd}[k] \delta h[k]$, $a'_{hd}[k+1] \approx a'_{hd}[k]$. The as-state estimator keeps a'_{hd} as a plain 5th KF state and *removes the var-ratio channel entirely* — no $\hat{a}'_{hdm}$, no β -EWMA. State $x = [\delta h_1, \delta h_2, \delta h_3, a_h, a'_{hd}]^T$.

Predict: $\hat{a}_h[k+1] = \hat{a}_h[k] + \hat{a}'_{hd}(\Delta h_d + (1 - \lambda_c) \delta \hat{h}_3)$, $\hat{a}'_{hd}[k+1] = \hat{a}'_{hd}[k]$ (integrator).

Update (KF only): $\hat{a}'_{hd}[k+1] = \hat{a}'_{hd}[k] + L_{51} e_{y1} + L_{52} e_{y2}$

With $H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$ ($H(\cdot, 5) = 0$), L_{5j} is a *cross-covariance* gain $\propto P_{5,4}^-$, built solely by $F_e(4, 5) = \Delta h_d$:

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh} & 0 \\ 0 & 0 & (1 - \lambda_c)\hat{a}'_{hd} & 1 + \hat{a}'_{hd}F_{dh} & \Delta h_d \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad Q_{55} = \sigma_{a''}^2 (\Delta h_d^2 + \sigma_{\delta h}^2).$$

Row-5 pole = 1 (pure integrator, cf. β in the co-update); a'_{hd} moves *only* through the L_e correction. $\sigma_{a''}^2 \sim (a_{nom}/R^2)^2$ is the sole curvature prior. This is the blue line: $\hat{a}_z$ on truth ($1.04\times$), tracking 26.5 nm; a'_{hd} underestimated in holds ($\Delta h_d \rightarrow 0$, cross-cov gain $\rightarrow 0$ — the hold-desert CRLB floor).

8.1 Why $\hat{a}_z$ jitters, and how to reduce it *a leak-free gain integrator anchored by a noisy a_{xm}*

The gain state a_h is a *leak-free integrator* ($F_e(4, 4) \approx 1$, no reversion): its only anchor is $y_2 = a_{xm}$, itself a finite-window variance inversion with relative std 0.37–0.54; and its predict injects thermal noise through $(1 - \lambda_c)\hat{a}'_{hd}\delta\hat{h}_3$. So even when the true gain is constant, $\hat{a}_z$ wanders $\sim 5\%$ around it:

segment	true a_z (nm/pN)	$\langle \hat{a}_z - a_z \rangle$ (nm/pN)	rel.
hold (far wall)	13.96	0.73	5.2%
descent	12.29	0.71	5.8%
osc (near wall)	9.23	0.51	5.5%
hold (near wall)	6.92	0.30	4.3%

The absolute wander tracks the gain (largest far from the wall, where a_{xm} 's noise $\propto a$ is larger). The balance is a Q_{44}/R_2 tension: raising Q_{44} tracks a changing gain (motion) but amplifies the a_{xm} noise; lowering it smooths but lags. With a_{xm} the *only* gain measurement, smooth-and-responsive cannot both hold — the residual $\sim 5\%$ is a CRLB floor.

How to reduce it. (i) Give the gain a reversion instead of a free integrator: the **kfmeas** a_d -state (deterministic predict $a_d + \hat{a}'_{hd}\Delta h_d$) cuts the *predict-side* wander $\sim 30\times$ ($2.74\% \rightarrow 0.09\%$), though the measurement-anchored jitter remains. (ii) Lengthen the a_{xm} window (smaller a_{cov}) to quiet the ruler, at the cost of lag. (iii) Break the floor only with *more information* — an independent gain channel or trajectory dither; tuning Q_{44}/R_2 alone merely slides along the smooth-vs-responsive trade-off.

8.2 If accurate $\hat{a}_z$ is the goal *a charter-compliant, derivable optimization path*

Under the charter ($c(\bar{h})$ unknown; only a_{nom}, R, T_s , the wall position, and the commanded trajectory are usable), two legal levers remain:

(a) *Structural — free and already derived.* Adopt the **kfmeas** a_d -state: its predict $a_d + \hat{a}'_{hd}\Delta h_d$ uses *only* the commanded Δh_d and the estimated $\hat{a}'_{hd}$, assumes no $c(\bar{h})$, and is the validated $\sim 30\times$ predict-wander reduction ($2.74\% \rightarrow 0.09\%$). This removes the leak-free random walk at zero information cost.

(b) *Information — to pass the CRLB floor.* The measurement-anchored jitter is set by a_{xm} being the *sole* gain channel; it falls only with an independent, c -free channel (a derived deterministic residual, unbiased slope 0.90–0.94) or trajectory dither (the trajectory is ours to design). Both are derivable and assume no wall-function form; tuning $a_{cov}/Q_{44}/R_2$ only trades jitter for lag and cannot pass the floor.

9. Appendix: the self-copy $\hat{a}'_{hdm} \equiv \hat{a}'_{hd}$ ($\kappa \approx 0$) *why the var-ratio carries no truth*

In a hold ($\Delta h_d = 0$) the gain predict reduces to $\hat{a}_h[k+1] = \hat{a}_h[k] + (1 - \lambda_c)\hat{a}'_{hd}\delta\hat{h}_3[k]$. Using the telescoping identity $(1 - \lambda_c)\delta\hat{h}_3[j] = \delta\hat{h}_3[j] - \delta\hat{h}_3[j+1]$ and summing from k_0 ,

$$\hat{a}_h[k] = \hat{a}_h[k_0] + \hat{a}'_{hd}(\delta\hat{h}_3[k_0] - \delta\hat{h}_3[k]) \implies \hat{a}_r[k] = -\hat{a}'_{hd}[k]\delta\hat{h}_r[k].$$

So $\hat{a}_r$ is built *from* $\hat{a}'_{hd}$, and the two variances are locked:

$$\hat{\sigma}_{\hat{a}_r}^2 = \hat{a}'_{hd}{}^2 \hat{\sigma}_{\delta\hat{h}}^2$$

$$\boxed{\hat{a}'_{hdm} = \sqrt{\hat{\sigma}_{\hat{a}_r}^2 / \hat{\sigma}_{\delta\hat{h}}^2} = \hat{a}'_{hd} \quad \text{for any true } a'_{hd}}$$

The readout echoes the estimate, independent of the truth; its fidelity is

$$\kappa := \frac{\partial \hat{a}'_{hdm}}{\partial a'_{\text{true}}} = 0.$$

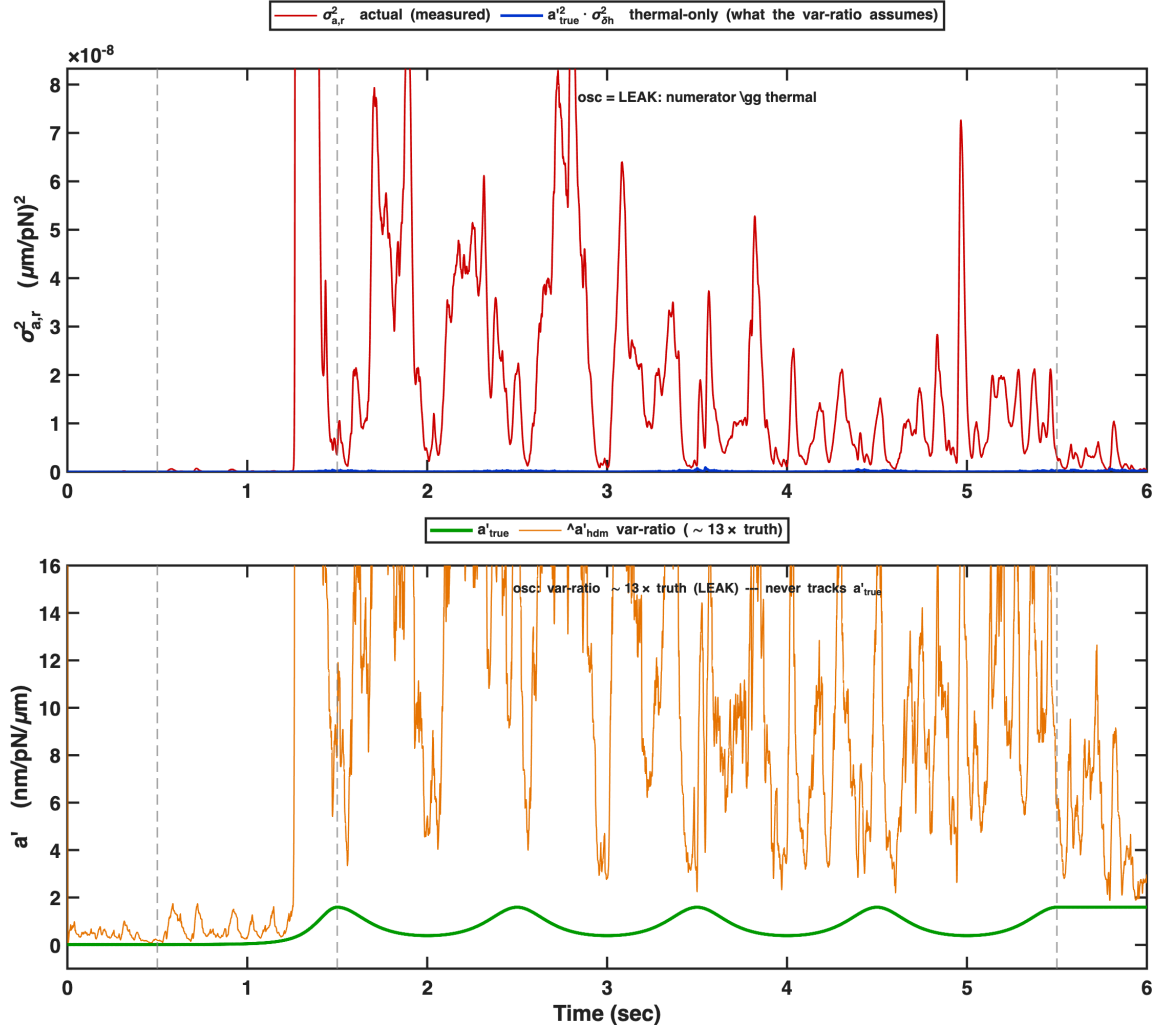
(The measured $\kappa \approx 0.03$ and the +2.5% sqrt-rectification bias $b_u \propto 1/\hat{a}'_{hd}$ come from the correction terms and finite EWMA windows; both are $\ll 1$, so the echo dominates.)

10. Appendix: the leak — the motion face of the self-copy *why the var-ratio is garbage in motion, and why the leak cannot be fixed*

In motion the gain estimate $\hat{a}_h$ swings to track the changing desired-height gain $a_{\text{det}}(t)$ (via the feedforward $\hat{a}'_{hd} \Delta h_d$). The mean-tracker $\bar{a}_h = \text{EWMA}(\hat{a}_h)$ is a low-pass; its complementary high-pass $\hat{a}_r = \hat{a}_h - \bar{a}_h$ does *not* remove this swing fully — it leaks a fraction

$$\lambda_{\text{leak}} \approx \frac{2\pi f T_s}{a_{\text{var}}} \approx 0.079$$

of it into $\hat{a}_r$. Since the deterministic swing ($\sim \text{nm/pN}$) dwarfs the thermal residual $a_r = -a'_{hd} \delta h$ ($\sim 0.04 \text{ nm/pN}$), even a 7.9% leak dominates: $\sigma_{\hat{a}_r}^2$ inflates far above the thermal $\hat{a}'_{hd}{}^2 \sigma_{\delta\hat{h}}^2$, so $\hat{a}'_{hdm} = \sqrt{\sigma_{\hat{a}_r}^2 / \sigma_{\delta\hat{h}}^2} \gg \hat{a}'_{hd}$.



Diagnostic var-ratio computed from a healthy as-state $\hat{a}'_{hd}$. Top: the actual numerator $\sigma_{a,r}^2$ (red) sits $\sim 170\times$ above the thermal-only value it should have (blue) during osc — the leak. Bottom: the var-ratio $\hat{a}'_{hdm}$ (orange) is $\sim 13\times$ truth (green) — garbage — during motion, and never tracks a'_{true} .

Why it cannot be fixed. The leaked swing is itself $\hat{a}'_{hd}$ -driven (the feedforward uses $\hat{a}'_{hd}$), so $\hat{a}'_{hdm}$ stays $\hat{a}'_{hd}$ -derived — the leak is the *motion* face of the same self-copy whose *hold* face is §9's echo. Removing the leak (faster mean-tracker, or subtracting the feedforward) only turns motion-garbage back into the hold echo $\hat{a}'_{hdm} \equiv \hat{a}'_{hd}$ — still no truth; the root ($\hat{a}_r$ built from $\hat{a}'_{hd}$) is untouched. The clean fix is not to repair the var-ratio but to drop it (pure *Le* / as-state, or 5state_aprime_kf_meas).